

SIXTH BIENNIAL REPORT

OF THE

STATE VETERINARIAN

OF CALIFORNIA

For the two years ending June 30, 1912

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REPORT OF STATE VETERINARIAN.

To His Excellency, HIRAM W. JOHNSON,
Governor of California.

SIR: I have the honor to submit herewith the Sixth Biennial Report of this department, showing, as near as it is possible to do so, the work accomplished during the sixty-second and sixty-third fiscal years. In this report I also desire to call your Excellency's attention to some very vital defects in our system of live stock sanitation, believing as I do that such defects can be corrected by adequate legislation with the consequent strengthening of this branch of the public service.

The people engaged in the live stock industry of this State are represented on the whole by a most highly representative citizenship. We have, of course, exceptions, as is the experience in all other official work; and while these exceptions in our case are very few, yet, to deal with them, it is necessary to have and enforce adequate laws for the protection of the masses.

The successful stockman studies all the angles of his business, not the least important of which is the prevention and extermination of animal diseases that might be prevalent. He realizes that in order to be successful such diseases must be kept out of his herd, or, if existent therein, adequate and sometimes drastic measures must be adopted, looking towards their eradication. He also knows that authority for the prevention and spread of these diseases must be invested in competent officials. So with these facts constantly before him, he coöperates with this department, without which coöperation, whether with or without good laws, our efforts to combat the animal infections would be very feeble indeed.

As an example of this intelligent coöperation between the live stock interests and this department, I would call your Excellency's attention to the almost complete extermination of Texas fever in cattle, and scabies in sheep, in this State, as well as the work that has been accomplished in exterminating glanders among horses and mules.

In order that your Excellency might have a more thorough understanding of the work that has been accomplished in this State, it is necessary for me to draw some comparisons. In 1904, cattle tick infestation, the cause of Texas fever in cattle, existed on many ranches in every county south of Merced and Monterey counties. The losses from this disease were so severe as to make the business of cattle raising in this permanently tick-infested area a very hazardous business. The cattle located in the territory north of this area were also constantly menaced, while at the same time the United States Government, and

northern and eastern states that were free from this disease, placed a ban on California cattle, which prevented their removal from this State. At the present time this tick infestation is confined to only a few ranches in San Diego County and the coast of San Luis Obispo County.

This active work of cattle tick eradication has been conducted for a number of years in all the southern states of this country. However, I am pleased to report that the records of the United States Department of Agriculture show that more square miles of tick-infested territory have been cleaned and removed from quarantine in California than the aggregate amount in the rest of the other states combined.

These results, I state frankly, are due to the intelligent coöperation of the stockmen with this department and officials of the United States Bureau of Animal Industry.

The history of sheep scabies and its control in this State is also an example of the results of coöperation, with the aid of the enforcement of adequate laws. This disease had been the bane of the sheep industry for many years. An inspection of the California flocks in 1908 showed that over 80 per cent of them were infected to a greater or less degree. During the same year, in coöperation with the United States Department of Agriculture, this department took up the work of systematically eradicating this disease. This work has been continued throughout the past two years, and while an occasional outbreak occurs in widely settled areas, the disease is under control and its ultimate extermination is assured.

During the thirty-seventh session of the legislature, the sheep men of California, thoroughly alarmed over the financial losses this disease had caused among their flocks, its constantly increasing ravages, with no adequate existent legal measures to cope with it, presented to the legislature a bill providing for the appointment of a Sheep Commission consisting of five wool growers, in addition to the State Veterinarian as an *ex officio* member, and investing in such board the power to adopt measures to exterminate scabies. This bill became a law, but never became operative, from the fact that the Governor refused to appoint this commission. It was believed at that time by the sheep men that the work of controlling and eradicating scabies should be in the hands of a board instead of investing such authority in the hands of the State Veterinarian alone, as, in the opinion of the sheep men, the task was too big for any one man. Seeing the necessity of some immediate action being taken after the Governor's failure to place this act into operation, the State Veterinarian interested the boards of supervisors in the various sheep-raising counties to appropriate funds with which to appoint sheep inspectors, who, in coöperation with this department and the United States Bureau of Animal Industry, might immediately proceed with this work. This was accomplished, and even under inadequate laws a good start was made in this work, until at the thirty-eighth session of the legislature laws were enacted that gave the State Veter-

inarian full authority to employ sheep inspectors, as well as adequate authority to exterminate scabies. It has since been proven to the satisfaction of the sheep men that the handling of all infectious diseases is better placed under the control of this department than by the creation of a board of commissioners.

The foregoing is not written for the purpose of calling your Excellency's attention to work accomplished by this department in the years prior to the last two, but, on the contrary, I am endeavoring to show you how a department like this may increase its usefulness and efficiency.

Two of the most insidious infectious diseases of animals that have ever existed are more or less prevalent in every state in the United States, as well as in other countries. I refer to glanders in horses and mules, and tuberculosis in cattle and hogs. California has for years and is still continuing to offer its tribute to these diseases. While this department has been exerting every available effort to combat them, and while we feel, so far as glanders is concerned, that our efforts have been rewarded by the eradication of many foci of infection, still we believe that great benefits can be accomplished to the State if our facilities for fighting these diseases could be strengthened.

During the past two years 1,252 head of horses and mules were destroyed on account of being infected with glanders. In the majority of instances we were enabled to dispose of these animals without any trouble, through the consent of the owners. In other instances it was found necessary to threaten the application of a very weak law if the owner refused to give his consent to the destruction of the animals. In other instances the owners positively refused to destroy such infected animals and appealed to the courts to restrain our actions. In San Francisco two horses were found visibly affected with this disease in the stable of B. Rosenberg. The balance of his horses, sixteen in number, were tested with the mallein and precipitation tests, and fourteen animals reacted positively. The owner of these animals applied to the Superior Court and secured a permanent injunction restraining the State Veterinarian, or any of his deputies, from killing or in any manner further interfering with these animals. Undoubtedly these animals have, since that time, disseminated this disease. In another large San Francisco stable an injunction was applied for nearly two years ago, and the case is still pending in the courts. The owners might just as well have been granted a permanent restraining order. Other similar legal tangles have occurred to interfere with the proper conduct of this work, but I call your attention to the foregoing only for the purpose of showing to you the inadequacy of our laws in the premises.

Glanders is a disease which has both an economical as well as a public health importance, as it is, of course, the cause of great losses to the horse and mule industry, and is communicable and usually fatal when contracted by human beings. It is, however, a disease that can be controlled by adequate laws properly applied; and, in the writer's

opinion, under such circumstances its control and ultimate eradication is an easier problem than the control and eradication of the fever ticks and sheep scabies, which the writer has elaborated upon in the foregoing part of this report.

The Attorney General, in an opinion rendered this office on May 26, 1909, in part advises with reference to our authority in connection with glanders, as follows:

"There is no statute in this State authorizing you, or any other public officer, to destroy animals afflicted with glanders as nuisances. But there is no doubt that under the statutes creating your office, and the amendments thereto, you have the right to enter the premises of any person in the State and examine animals for the purpose of discovering contagious diseases; and if you find stock affected with glanders, your recourse in the present state of the law is to notify the owner of the presence of the disease, call his attention to section 402b of the Penal Code, and, in case he fails to comply therewith, cause a criminal action to be brought against him."

I must again refer back to cattle tick infestation and sheep scabies, which are now nearly extinct in this State.

In the work of exterminating these two pests the State Veterinarian was aided by adequate legislation that authorized him to seize and disinfect cattle and sheep after the owner had been officially notified and failed to do so. Under such laws, and the proper enforcement of them, such work has progressed with rapidity.

Laws that could be applied to glanders and other diseases in the same manner, giving the state officials proper authority, would, in my opinion, be productive of the same results.

In the handling of an outbreak of glanders this department has always insisted upon the immediate slaughter of all animals exhibiting typical clinical symptoms, about which there could be no question of error in diagnosis. We have done this under authority of an earlier opinion in the premises from the Attorney General in which it is stated that the proper official has the authority to destroy an animal which is affected with an incurable communicable disease when such a disease is a menace to public health and safety. When an official does this, the opinion goes on to state, he is personally responsible to the owner of the animal in case he has made an erroneous diagnosis.

The chief difficulty in exterminating an outbreak of glanders is not in the destruction of the clinically affected animals, but is, on the contrary, the handling of those cases where the lesions are confined to the internal organs. The detection of animals so affected is brought about by means of the application of the recognized biological tests for this disease.

The procedure adopted by this department in exterminating glanders is to destroy the clinically affected animals, clean and disinfect the premises and then proceed to test the remaining contact horses and mules. Almost invariably, as a result of this test, one or more positive reactors are discovered, showing that such animals are affected with the internal form of glanders. Many of such animals, if not properly disposed of, will, in time, develop clinical symptoms of the disease, and in

all events a certain number of them will disseminate glanders to other animals, even if they never develop into clinical cases.

Section 402*b* of the Penal Code of this State provides:

"Every animal having glanders or farcy shall at once be deprived of life by the owner or person having charge thereof, upon discovery or knowledge of its condition; and any such owner or person omitting or refusing to comply with the provisions of this section shall be guilty of a misdemeanor."

This law makes no distinction between the clinically affected animal and the one suffering with internal lesions alone. For this reason, and also from the fact that the disease is impossible to control if such occult cases are not properly disposed of, this department has insisted, as far as the courts would allow, on the slaughter of all typically reacting animals.

This is a sanitary police policy that is universally recognized, but there are economical objections to this policy by horse owners, which, under the existing laws of this State, are, in my opinion, founded not without good reasons.

Very few states demand the slaughter or proper segregation of animals which react to the recognized tests for glanders, but, in my opinion, when this is not done, the killing of clinically affected animals alone is only palliative in effect and is productive of no lasting benefits, as far as the eradication of this disease is concerned.

In several states such reacting animals are destroyed, their value appraised, and their owners indemnified by the State for at least a part of their value, usually two thirds.

I believe legislation can be adopted in California that will obviate the necessity of destroying animals which react to the tests, and that such animals may be allowed to live and work within certain police restrictions without endangering healthy horses and mules. It would be very practicable, in my opinion, to place such animals under a provisional quarantine, directly under the supervision of this department.

In accordance with the foregoing views, I would respectfully recommend that the next legislature enact legislation, giving the State Veterinarian the legal authority to destroy all animals which are visibly affected with glanders. Furthermore, I would recommend that the State Veterinarian be legally authorized to test, with the recognized tests for glanders, all horses and mules that have been in contact with such visibly affected glandered animals, and that all such horses and mules which positively react to such tests be immediately segregated from the balance of the animals which give negative reactions. The State Veterinarian should be further authorized to brand such reactors in such manner as to insure their identities for the balance of their lives. It should be further provided that the owner of these reacting animals be allowed to use them, provided they are not permitted to come into contact with healthy animals. This law should explicitly provide what contact should consist of, and, in my opinion, it should be adequately covered by the following:

All condemned animals must be stabled and worked in such a manner

that they are not brought into direct contact with other animals. They must be watered separately and from different receptacles from which other animals are watered.

They must not be fed from the same racks or mangers which other animals are fed from.

While on the road or other public highway they must not be watered from public troughs or hitched at public hitching places.

An inspection must be made at regular intervals of such animals, and in case any of them are found to have developed external symptoms, the State Veterinarian should be authorized to destroy same.

The owner of such animals should not be allowed to dispose of them in any manner without first notifying and obtaining the approval of the State Veterinarian.

It should be further provided in such a law that if the owner or controller of these animals infracts any of the provisions of this quarantine, in such case the State Veterinarian should be legally authorized to destroy the animals in question.

Provision should also be made for an early court hearing in case proceedings be taken by any owner of such condemned animals to contest any provision of the law, and while such hearings are being held the animals in question should be placed in the custody of the State Veterinarian.

Much of what I have said about glanders in horses and mules may be applied to tuberculosis in cattle and hogs, inasmuch as a system of control of this disease might be worked out.

Tuberculosis is very prevalent among our dairy herds, and, with the exception of a herd scattered here and there in various localities, there is no doubt but that this disease is on the increase.

The extent to which this disease has spread to hogs which are raised and fed in connection with dairies is alarming. It frequently occurs that the greater percentage of hogs in a shipment to an abattoir are found tuberculous by the inspectors on autopsy.

This disease is also increasing on the ranges. Herds of cattle in California, known to have been free from this disease a few years ago, are now infected, and in some instances the disease has spread so rapidly among the members of the herd that the situation is indeed serious.

The opinion prevails that our beef herds, on account of their open-air existence, are very resistant and consequently free from tuberculosis. This is a very erroneous idea, for this department is cognizant of the existence of tuberculosis in some of our range herds, the members of which have never been housed during their entire lives.

Tuberculosis becomes seeded in range herds by carelessness in bringing on to the range animals which are not known to be positively free from this disease. From such animals the disease spreads to other individuals in the herd. The chief offenders are the dairy cows and breeding bulls.

The subject of tuberculosis in animals is an extremely important one, involving as it does an economical importance to the live stock owner

as well as its relationship to human health. No doubt exists to-day concerning the transmission of this disease to human beings by infected cow's milk and meat; in fact, many investigators ascribe infected milk as the main source of human infection.

The chief obstacle and objection to the adoption of measures looking towards the control of bovine tuberculosis appears to come from resignation, due to its general prevalence and insidious nature. It is true that in a number of localities in this and other states measures were adopted with a view of exterminating this disease, but in many instances these measures proved abortive and were discontinued. This, in my opinion, was due to the fact that proper and sufficient consideration was not given to the big subject involved. At a meeting of the California Live Stock Sanitary Association held in Los Angeles in December, 1911, the following resolution was adopted:

WHEREAS, Tuberculosis prevails and is increasing to an alarming extent in this State among domesticated animals, more especially cattle and hogs; and

WHEREAS, It has been abundantly proven that said disease is transmissible from domestic animals to human beings through the media of milk and meat, more especially to young human beings, whose sole diet is frequently obtained from tuberculous dairy cows; and

WHEREAS, From an economic standpoint the existence of said disease among domestic animals is a source of enormous financial loss to dairymen, meat producers, breeders, and those engaged in other agricultural pursuits; therefore, be it

Resolved, That this Association instruct its president and secretary to request the Governor of the State of California to appoint a commission whose duty it will be to devise feasible and practical plans for the control and eradication of tuberculosis among domestic animals, said plans to be submitted to the State Legislature for legislative enactment following the Governor's approval of the same. Such commission to consist of seven members as follows:

Two veterinarians who are and have for the past five years been actively engaged in sanitary work.

One dairyman who is and has for the past five years been actively engaged in the production of milk and milk products.

One stockman who is and has for the past five years been actively engaged in the breeding of stock for meat purposes.

One physician who is and has for the past five years been actively engaged in sanitary work.

One hog raiser who is and has for the past five years been actively engaged in the breeding and raising of hogs; and

One person in no way connected with the above mentioned industries or professions who will represent the consuming public.

The members of said Commission to serve without compensation.

I believe it would be better if such a resolution could be presented to and adopted at the next session of our legislature, empowering your Excellency to appoint a commission to study this subject with a view of presenting a report and legislation to the following legislature.

I believe the personnel of such a commission should either be changed or be enlarged so as to provide representation thereon from the California Federation of Women's Clubs, as this subject has been very thoroughly discussed and is well understood by the members of that body.

At the thirty-ninth session of the legislature, legislation was enacted which provided that all dairy and breeding cattle offered for shipment into this State must be inspected and tested with tuberculin prior to

shipment. The intent of this law was not to subject strictly range cattle to the tuberculin test. However, the wording of the act is ambiguous in this respect, and I would recommend that it be amended so as to specifically exempt strictly range cattle. At the same session of the legislature a similar law was enacted providing for the inspection and mallein testing of all horses, mules and asses before shipment into California. An exemption to the testing of unbroken animals should have been provided in this law, and I would therefore recommend that such an amendment be enacted during the next session of the legislature.

Confusion and annoyance very often occurs to live stock shippers, as well as railroad agents, by the multiplicity of county ordinances providing for the inspection of live stock before shipment into their respective counties. The chief difficulty appears to be in the differences in such regulations. It would simplify matters, and better facilitate the movements of live stock within the State if uniform ordinances could be adopted by the various counties; or, better still, if the intrastate movements of live stock could be fully supervised by this office, under the provisions of an act of the legislature, if such would be constitutional. Such a law would meet with the approval of the shipper as well as the transportation companies.

In the general conduct of the work of this department four statutory officers are employed, viz.: the State Veterinarian, Assistant State Veterinarian, who is located in Los Angeles; a Deputy State Veterinarian, and a clerk. Another veterinarian is employed and his salary is paid out of the appropriation for traveling and contingent expenses. A live stock inspector, who is a qualified veterinarian, is also employed in tick eradication work. His salary is also paid out of the appropriation for traveling and contingent expenses. Sheep inspectors are employed principally during the spring and fall months, and their salaries are paid out of the appropriation of \$10,000 allowed by the thirty-ninth session of the legislature for the payment of inspectors to eradicate scabies in sheep. The work of the veterinarians consumes their entire time, and keeps them from their homes most of the time. This work, in order to be successful, requires technical skill, and in many instances, tact.

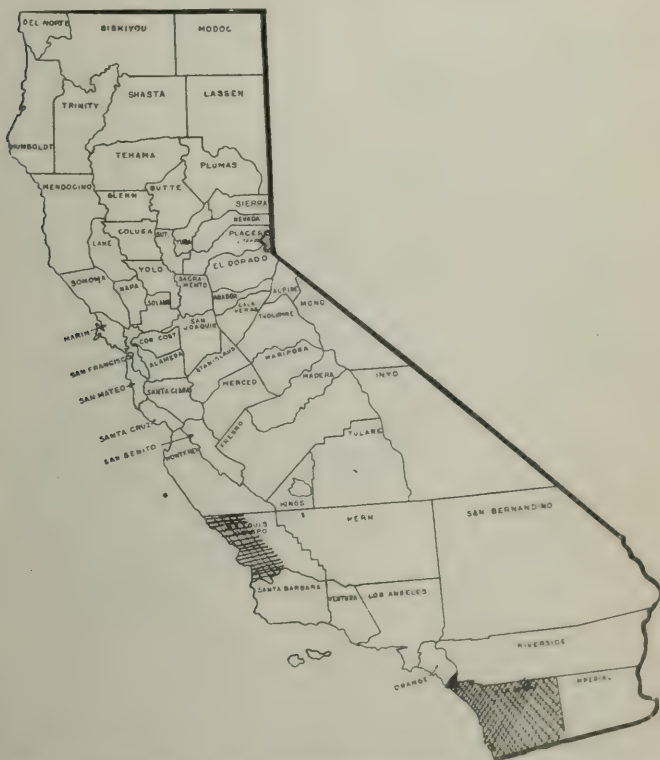
The statutory salary of the Deputy State Veterinarian is \$1,800.00 per annum. This, I believe, is insufficient, and would therefore recommend an increase of such salary to \$2,400.00 per annum.

I take pleasure in calling your Excellency's attention to the most satisfactory manner in which the work of this department has been carried on in the southern end of the State by the Assistant State Veterinarian, Dr. W. E. D. Morrison; also to the efficient manner in which Deputy State Veterinarian Dr. J. P. Iverson has conducted his work.

SOUTHERN CATTLE FEVER.

During the past two fiscal years the following areas of territory have been freed from cattle ticks and removed from federal and state quarantine:

October 16, 1911, by proclamation, your Excellency released all of Santa Barbara County.



Dark areas represent sections infested with Southern cattle fever ticks and quarantined for same, June 30, 1912.

March 7, 1912, by proclamation, your Excellency removed from quarantine all of Orange County except a small area in the extreme southwestern part of said county.

This leaves at the end of June 30, 1912, only San Diego County, that part of San Luis Obispo County lying west of the summit of the Santa

Lucia Range of mountains, and a small area in the southwestern part of Orange County, under provisional quarantine. However, the work of tick eradication is so well in hand in the above mentioned sections



Dark area represents the counties infested with Southern cattle fever ticks and quarantined for same, in 1904.

that cattle are allowed to move anywhere after they have passed inspection. During the two years just passed there has not been any extension of tick infestation to uninfested ranches, and the losses of cattle from Texas fever have been nil. From present indications the balance of Orange County, as well as the balance of San Luis Obispo County, will be freed from ticks and ready for removal from quarantine restrictions next year.

HOG CHOLERA.

Cholera in hogs has been reported and investigated by this department during the past two years in the following counties: Amador, Butte, Contra Costa, Fresno, Imperial, Kings, Madera, Merced, Napa, Orange, Placer, Santa Clara, San Diego, San Francisco, Solano, Stanislaus, San Joaquin, Sutter, Tehama, Tulare, and Yuba.

Prior to enactment by the last legislature of the law providing for the preparation of anti-hog-cholera serum by the State Experiment Station, and its distribution at cost of production, this department could only handle this disease by general sanitary and quarantine measures. The ease by which this disease can be carried from its original focus by intermediate bearers rendered our efforts, in many instances, hopeless. That the sera will prove of great value as an adjunct in assisting us to control this plague, is evidenced by reports that have been received from hog raisers who have taken advantage of its use. However, in order that its value to the State might be extended, hog raisers should not wait for the disease to become evident in their herds, but, on the contrary, the hogs on all ranches in the neighborhood where the disease exists should be immunized. The officers of this department have investigated in a number of instances reports of deaths among hogs from cholera, and after careful investigation we have found fatalities occurring from other causes.

ANTHRAX.

Fifty-five outbreaks of anthrax were reported to this department and investigated during the past two years. These outbreaks occurred among cattle, sheep, horses, mules and swine in the following counties: Sacramento, Merced, Nevada, San Benito, San Mateo, Tulare, Kings, Fresno, San Joaquin, Stanislaus, Mono, Butte, Solano, Napa, Contra Costa, Marin, Los Angeles, Colusa, Madera, and Glenn. Undoubtedly other outbreaks of this disease occurred on the ranges without the knowledge of this department.

The control of anthrax is a very perplexing problem. This disease has existed on a number of ranges in this State for many years. This is especially the case in cattle ranges in sections of the State along the Sacramento and San Joaquin rivers. Unfortunately the infection of anthrax, when once seeded on one of these ranges, persists more or less indefinitely, and when animals are fed off these infected pastures fatalities occur year after year. Under such circumstances the disease is, of course, difficult to control. This department has no legal authority to prevent an owner from turning his cattle out on an infected range. All we can do in the premises is to quarantine the ranch and order proper disposition of carcasses. Preventive vaccination is often resorted to, but in many instances without success. This may be attributed to the fact that among the anthrax vaccines on the market some are impotent, while in some cases where potent vaccines are used failures are due to their improper use. This is well evidenced in the many cases where cattle are vaccinated during the progress of the disease in the herd. After vaccination such cattle are immediately turned out again on to the infected pasture, the owner of such cattle expecting immunity to be conferred immediately. When reliable vaccine is properly used, the deaths from anthrax are lessened or stopped entirely. However, no matter how reliable the vaccine, immunity is not set up immediately after vaccination. On the contrary, when animals are vaccinated during the existence of the disease in the herd, the vaccinated animals should not again be turned out on to the infected range until sufficient time has elapsed for the production of immunity. Several weeks will suffice. Of course, a better sanitary procedure is to vaccinate the animals some time before the disease usually becomes evident. The sale of anthrax vaccine should be regulated by the State in order that live stock men might be protected from purchasing and using unreliable anthrax vaccines.

The great danger to human beings in contracting anthrax, especially when the disease exists in a dairy herd, is not overlooked by the inspectors of this department when handling an outbreak. A number of such outbreaks have occurred in dairy herds during the past two years, and in such cases the use and sale of the milk and its products are prohibited until such a time as we are satisfied that the disease is under control.

RABIES.

In the last biennial report of this department attention was called to the fact that unless known methods of control of this disease were properly enforced, that it would continue to spread. During the two years just past, California has probably suffered with just as severe and extensive an outbreak of this disease as has occurred in any state in this country. The disease has spread with considerable rapidity from the southern section of the State through the San Joaquin Valley counties, until, at present writing, San Francisco and neighboring counties are included in the area of infected territory. Under the present system of control, rabies will undoubtedly spread over the entire State.

The health officials in some of the southern and San Joaquin Valley cities, as well as the officials in San Francisco, must be given great credit for the manner in which they have striven to combat this disease. However, it is very unfortunate that the State has been unable, other than in an educational and advisory manner, to assist them and make their work complete. No matter what efforts a municipality may resort to in order to control and exterminate rabies, such efforts are often unavailing unless the officials of neighboring infected territory do likewise.

While the reports of cases of rabies affecting the lower animals received at this office do not anywhere near represent the extent of the infection within this State, the following table, showing the cases reported to this office, will, however, give you some idea of the extent of territory involved:

Reports of Rabies—July 1, 1910, to June 30, 1912.

Counties	Dogs.	Horses.	Mules.	Cows.	Calves	Hogs.	Goats.	Total.
Los Angeles	42	1	-----	1	-----	-----	-----	44
Fresno	79	2	-----	-----	-----	-----	-----	81
Merced	22	-----	-----	3	-----	4	-----	29
Ventura	1	-----	-----	-----	-----	-----	-----	1
Kings	2	-----	-----	-----	-----	-----	-----	2
Riverside	5	-----	-----	-----	-----	-----	-----	5
San Luis Obispo	2	-----	-----	1	-----	-----	-----	3
Tulare	22	-----	-----	-----	-----	1	1	25
San Francisco	62	-----	-----	-----	2	-----	-----	64
Kern	18	2	1	-----	-----	-----	-----	21
Orange	5	-----	-----	-----	-----	-----	-----	5
Santa Barbara	2	-----	-----	-----	-----	-----	-----	2
Santa Clara	2	-----	-----	-----	-----	-----	-----	2
Madera	3	-----	-----	-----	-----	-----	-----	3
Sacramento	-----	-----	-----	2	-----	-----	-----	2
Stanislaus	3	1	-----	3	-----	-----	-----	7
San Mateo	2	-----	-----	-----	-----	-----	-----	2
Alameda	1	-----	-----	-----	-----	-----	-----	1
	273	7	1	10	2	5	1	299

GLANDERS.

The efforts of this department to eradicate this disease have been continued with vigor throughout the past two fiscal years. In all, 635 reports of glanders infection were investigated, and 1,252 head of horses and mules were destroyed. This is a total of 450 more horses and mules than were destroyed during the previous two fiscal years, as indicated in my biennial report for that period. From the increase in the number of destroyed animals it should not be inferred that glanders is on the increase in this State, but, on the contrary, the efforts of this department in the work of exterminating this disease have been doubled and many old foci of disease, hitherto hidden, have been discovered. This is well exemplified by the discovery of diseased animals, which, upon tracing their history, we have ascertained had been affected with chronic glanders for several years. We have endeavored in every instance where a case of glanders has been reported to trace the ownership of the animal back as far as it is possible to do so, and in this manner we have been enabled, in many instances, to run on to a trail of infection among other animals that was left behind in the change of ownership and location of the chronically affected animal.

It will be noted in the summary that accompanies this report that of the 1,252 animals which were destroyed, 612 were killed without a test. All of such animals exhibited the well-defined clinical symptoms of glanders, consequently the testing of them was unnecessary. In all 3,647 animals, the majority of which exhibited no external symptoms of glanders, were found to have been in contact with the 612 clinical cases, and such animals were quarantined and tested with the recognized biological tests in order to determine if any of them were affected with the internal form of the disease. Of these 3,647 contact animals, 640 gave typical reactions to the tests and were also destroyed; 2,964 animals successfully passed the tests and were released from quarantine. The majority of tests employed were with mallein. Nearly all of this mallein was furnished to this office without cost by the Pathological Laboratory of the United States Bureau of Animal Industry, Washington, D. C. The precipitation test was used on several hundred head of animals conjointly with the mallein test. Early indications lead us to believe that the arduous mallein test might be discarded for this simpler test. However, in our opinion, the precipitation test was not as reliable as the mallein test and it was therefore discarded.

During the last fiscal year a sera test, known as the complement-fixation test, was used on several hundred head of horses. Nearly all of such tests were conducted for us in the Pathological Laboratory of the United States Bureau of Animal Industry, Washington, D. C., through the courtesy of the Chief of the Bureau. Check mallein tests were also made on all of such animals with a view of comparing the two tests. The complement-fixation test for the detection of glanders is similar in principle to the Wasserman test for syphilis in man. The opinion prevails in scientific work that this test is possibly the most

accurate method known for detecting glanders, and it has been made the official test in several states in Germany as well as in parts of this country. Our experience with it would bear out this view. It is more accurate than the mallein test, and also possesses many other advantages. In the use of it there is no necessity to keep the animals away from their work, as it is necessary to do in applying the mallein test. The inspector is enabled to do his testing with more dispatch and less danger to himself. However, the final work in connection with the complement-fixation test must be done in the laboratory, so if the State wishes to take advantage of it, laboratory facilities, as well as a laboratory worker, must be supplied.

The following summary shows by counties the number of animals destroyed without testing; number tested and passed; the number which indefinitely reacted on the first test and consequently were retested; number which reacted to the test and were destroyed, and the total destroyed in each county:

Counties.	Tested.	Destroyed without test.	Tested and passed.	Held for retest.	Tested and destroyed.	Total destroyed.	Reactors not disposed of.
Alameda	103	47	64		9	56	30
Butte	4	1	1		3	4	
Calaveras		1			1	1	
Contra Costa	40	3	35	1	4	7	
Colusa	10	1	2		2	3	6
El Dorado						2	
Fresno	55	54	42		13	67	
Glenn	53	2	45	1	7	9	
Humboldt	17		16		1	1	
Imperial	13	7	11		2	9	
Kern	96	130	71	4	21	151	
Kings	44	19	21		23	42	
Lassen	34	4	34			4	
Los Angeles	481	71	336	15	127	198	3
Madera	4		3		1	1	
Marin		2				2	
Mariposa	1		1				
Mendocino	10	1	9		1	2	
Merced	48	12	47		1	13	
Monterey	78	1	73		5	6	
Napa	20	1	16		4	8	
Orange	96	14	72		24	28	
Placer	202	19	162		40	59	
Riverside	31	1	22	1	8	9	
Sacramento	57	13	34	3	20	33	
San Benito	1	2			1	3	
San Bernardino	174	15	91		83	98	
Santa Clara	121	12	89	1	31	43	
San Diego	104	3	73		31	34	
San Francisco	511	81	433	13	24	105	41
San Joaquin	318	32	335		23	55	60
San Luis Obispo	23	14	19		4	18	
San Mateo	18	7	11		7	14	
Sierra		5				5	
Solano	42	1	31		11	12	
Sonoma		1				1	
Stanislaus	8	7	4		4	11	
Sutter	18		18				
Shasta	2		1		1	1	
Tulare	358	18	184	1	70	88	3
Monterey	272	4	239		33	37	
Yuba	280	1	276	3	1	2	
	3,647	612	2,821	43	640	1,252	143

TUBERCULOSIS.

One hundred and fifty-seven herds of cattle were tested for tuberculosis by official veterinarians with tuberculin supplied from this office. One hundred and six of these herds were tested by this department. All of the tuberculin was furnished free to this department by the Pathological Laboratory of the United States Bureau of Animal Industry, Washington, D. C.

Seeing the necessity of doing something for owners of dairy herds who desired advice and assistance in ridding such herds of tuberculous animals, and thereafter maintaining healthy herds, this department, late in 1911, decided to test herds without any expense to the owners, under the provisions of an agreement similar to the following:

STATE OF CALIFORNIA,
OFFICE OF STATE VETERINARIAN,
SACRAMENTO, CALIFORNIA.

AGREEMENT.

In consideration of the testing of my herd of cattle by the State Veterinarian of the State of California, and the assistance of said State Veterinarian in enabling me to produce and maintain a herd of cattle free from contamination of tuberculosis, I, _____, of _____, owner of said herd of cattle, comprising _____ do agree as follows:

1. I will cause all animals which react to the tuberculin test, and which also show other marked symptoms of tuberculosis, to be slaughtered within a reasonable time under the United States meat inspection regulations, and I will cause the carcasses of said animals to be disposed of according to the meat-inspection regulations of the Bureau of Animal Industry, based upon the lesions found upon inspection.

2. I will cause all animals which react to the tuberculin test, but which show no other evidence of tuberculosis, either to be slaughtered and disposed of as herein provided for animals which show also other evidence of tuberculosis, or I will cause such animals to be removed from the herd and portion of the farm upon which the healthy animals of the herd are maintained, and I will cause the diseased animals to be segregated from the healthy animals, and thereafter they shall remain so segregated.

3. In all cases where the milk from such segregated reacting cows is to be used for any purpose whatever I will cause the said milk to be sterilized.

4. Segregated reacting bulls may be used for breeding, provided they are held on leash and are not permitted to leave the premises reserved for their use, and provided the healthy cows bred to such bulls are not unduly exposed to infected premises or to other diseased cattle.

5. I will cause the young from segregated reacting animals to be removed from their mothers at birth, and will not permit the said young to suck their mothers.

6. Any part of my premises contaminated by reacting animals will be submitted by me to a thorough disinfection under the direction or supervision of the State Veterinarian.

7. All cattle owned by me, both healthy and tuberculous, I will mark, or allow to be marked, in such manner as to enable their identity to be retained, and I will not change the location of or slaughter any tuberculous cattle except after due and timely notification to the State Veterinarian, stating the exact nature of the change of location, or the exact date, name, and address of the official establishment at which the animal or animals are to be slaughtered.

8. I will add no cattle to the said herd which have not passed a tuberculin test administered by a state or county inspector.

9. I will comply with all reasonable sanitary measures which are indicated by the proper officials of the State and county wherein my herd is located.

In Witness Whereof, I have signed this agreement this ___ day of _____
one thousand nine hundred _____

Witness : _____ Owner of the _____ herd.
Address _____

This work could only be undertaken when the employees of this department were not engaged in other work. For this reason we did not encourage it as much as we would have liked to do, and in a number of instances it was necessary to refuse requests as our time was fully occupied in handling other infectious diseases.

As will be noted in the following summary, 3,142 head of cattle were tested for tuberculosis. Twenty head were destroyed on account of visible symptoms of this disease. Out of the 3,142 head tested, 598 reacted to the test and were condemned and segregated from the balance, 2,452, which successfully passed. Ninety-two animals gave atypical reactions and are held for retesting.

Summary of Tuberculin Tests.

Counties.	Tested.	Destroyed without test.	Tested and passed.	Held for retest.	Reactors condemned.
Monterey	25		25		
Fresno	136	12	84	2	50
Kern	251		188		63
Kings		5			
Merced	14		10	3	1
Marin		1			
Mendocino	139		50	13	76
Napa	378	1	353	3	22
Riverside	846		734	14	98
Placer	3		2		1
Stanislaus	91		48		7
Sacramento	135		57	24	54
Solano	38		32		6
Sonoma	244		184	3	57
Santa Clara	73		10	2	61
San Joaquin	110	1	46	5	59
Tehama	18		18		
Tulare	154		154		
Los Angeles	412		346	23	43
San Bernardino	66		66		
Yolo	9		9		
	3142	20	2452	92	598

CEREBRO-SPINAL MENINGITIS (So-called).

Forage or mould poisoning, or so-called cerebro-spinal meningitis in horses and mules, has continued at intervals to cause severe losses in spite of the publicity we have given to this affection and its easy method of control. A number of very severe outbreaks of this trouble have been attended by officials of this department, one outbreak especially being attendant with severe loss. This occurred on a ranch in Butte County, and at the time my attention was called to it twelve animals were affected and a number had already died. A complete change to bright, clean fodder abruptly stops this trouble.

The disease is caused by eating fodder that has been invaded by a toxic fungus. It always occurs when the weather is cloudy, damp and warm. Such conditions are conducive to the growth of moulds. Such moulds may contaminate cut feed, as well as uncut forage.

Outbreaks of this disease were investigated in the following counties: San Luis Obispo, San Diego, San Joaquin, Sonoma, Lassen, Los Angeles, Humboldt, Glenn, Santa Clara, Fresno, and Butte.

SYMPTOMATIC ANTHRAX (Blackleg).

Fatalities among young cattle due to blackleg occurred in the following counties: Sonoma, Fresno, Yuba, Sutter, Ventura, Santa Barbara, San Luis Obispo, Merced, San Joaquin, Placer, Glenn, Tulare, and San Benito. While this disease is more or less generally prevalent on the range, it is easily controlled by vaccination. There is no excuse for a cattleman to suffer severe losses from it, for if he vaccinates his young cattle early enough in the year this disease can easily be prevented. The activities of this department in connection with it were in the nature of investigating the reports of deaths among cattle in order to establish a diagnosis, and when the disease proved to be blackleg, causing the cremation or deep burial of the carcasses, and recommending immediate vaccination of the balance of the young members of the herd.

SHEEP SCABIES.

Coöperative work with the United States Bureau of Animal Industry in the work of eradicating scabies in our flocks has been continued by this department during the past two years. While this disease has been under control for several years, yet systematic inspections must be continually made of all California sheep in order to detect as early as possible any infected bands. On account of the manner in which California sheep are run, and their migration to and from summer and winter ranges, one band of infected sheep, if not detected in time, will often scatter the infection to other bands. This has occurred in a number of instances, but it cannot, however, be ascribed to carelessness on the part of the inspectors. Ofttimes the parasitic cause of this disease will remain dormant in a band of sheep, and at the time the inspection is made the sheep are apparently clean, only to break out later under favorable conditions. In such cases, if the sheep have been trailed, the disease is often spread to other bands.

MISCELLANEOUS INVESTIGATIONS.

Sickness and deaths among domestic animals, from causes other than communicable diseases, are more or less constantly occurring in different localities in this State. Often an investigation is necessary in order to eliminate the possibility of an infection being the cause, while in all instances the advice and assistance of this department are sought for in order to afford relief. Improper care and feeding of animals was, in many instances, the cause of such sickness, and when the advice and instructions of the inspectors were followed, the trouble ceased.

In a number of instances investigations were made on the range to determine the cause of death among live stock where poisonous plants were suspected. Such investigations were made in Los Angeles, Plumas, Lake, Humboldt, Amador, and El Dorado counties. In such instances evidences of poisoning of live stock occurred from eating loco-weed (*Astragalus mollissimus*), different varieties of larkspur (*Delphinium*), lupines, and poisonous hemlock (*Conium maculatum*). The establishment of a diagnosis in such cases could, of course, be based only on the symptoms presented by the sick animal, results of autopsies, history of the trouble, and the existence of the particular poisonous plants in the neighborhood. Lack of finances and facilities for the making of thorough investigations of such poisonings on the range rendered some of this work inconclusive.

NEVADA QUARANTINE AGAINST CALIFORNIA CATTLE.

On May 14, 1912, I was advised by your Excellency of the issuance of a quarantine by the Governor of Nevada against cattle from all that part of the State of California situated south of the Mount Diablo base line. Following is a copy of this quarantine proclamation:

STATE OF NEVADA,
EXECUTIVE DEPARTMENT,
CARSON CITY.

A PROCLAMATION BY THE GOVERNOR.

WHEREAS, It has come to my knowledge through a petition of stockmen of western Nevada, praying for a proclamation of quarantine against a portion of the State of California satisfactorily appearing to be infected with the following contagious diseases prevailing among the live stock of such portion of said state, to wit: Texas, splenic or tick fever, tuberculosis, blackleg and anthrax, and

WHEREAS, It appears that their shipment into and presence upon the ranges of this State is a menace to the cattle industry of the State of Nevada, and

WHEREAS, In order to protect one of the principal industries of this State it is necessary that said disease be not allowed to be spread among our cattle and on our ranges,

Now, therefore, I, Tasker L. Oddie, Governor of the State of Nevada, do, by authority of law in me vested, hereby prohibit the shipment or importation into this State of cattle from that portion of California lying south of Mount Diablo base line, hereby declared to be an infected district and quarantined against, which are diseased or infected with any of the contagious diseases in this proclamation above mentioned, or which have been exposed to any of said contagious diseases; and do hereby urge upon all officers, whose duty it is, or may become, to rigidly enforce said quarantine:

Provided, That said prohibition shall not apply to any cattle being transported through and beyond the limits of this State by any railroad company by permit or authority granted by any agent or inspector acting under the instruction and authority of the United States Government.

Given under my hand and the Great Seal of the State of Nevada, at the Capitol in Carson City, this fourteenth day of May, in the year of Our Lord one thousand nine hundred and twelve.

TASKER L. ODDIE, Governor.

[SEAL.]

By the Governor:

GEORGE BRODIGAN, Secretary of State.

About the time this quarantine went into effect one of the largest cattle outfits in California had leased some cattle range in Mono County in this State and was contemplating moving a number of trainloads of cattle from their ranges in the San Joaquin Valley, where feed was short, to their range in Mono County. In order to do this, it would be necessary to transport the cattle by railroad to Wabuska, Nevada, and then make the drive overland into Mono County. It appears from the results of my investigation that the quarantine was issued in order to forestall the movements of these cattle.

The quarantine proclamation specifically states that cattle in the territory quarantined appear to be infected with Texas, splenic, or tick fever, tuberculosis, blackleg and anthrax. The quarantine described the same area in California that was quarantined for Texas tick infestation by the United States Department of Agriculture twelve years ago.

Since that time tick infestation has been completely eradicated in the San Joaquin Valley.

Anthrax and blackleg do exist to a limited extent among cattle in the San Joaquin Valley, just as they do in every state where there are large cattle ranges, including the State of Nevada. However, the incongruous part of it is that California has a system of live stock sanitary control, while the State of Nevada has no such system nor officials.

On May 23d I called on Governor Oddie at Carson City and discussed the advisability of an early revocation of this quarantine. The Governor informed me that he had requested United States Senators Newlands and Nixon of Nevada to take the matter up with the Secretary of the United States Department of Agriculture for the purpose of ascertaining through his officials if the continuance of such a quarantine was necessary to protect the health of Nevada live stock.

On my return to Sacramento, the following communication was forwarded to the Chief of the Bureau of Animal Industry, Washington, D. C.:

MAY 25, 1912.

Chief, Bureau Animal Industry, U. S. Department of Agriculture, Washington, D. C.

Sir: Relative to my telegraphic communication to you on the subject of the State of Nevada quarantining cattle from entering Nevada from all that part of California situated south of the Mount Diablo base line in this State, beg to further advise you as follows:

I have just returned from the State of Nevada, where I have been in conference with Governor Oddie of that State in relation to this matter.

The Governor informed me that in compliance with a request from this office made about a week ago, he had taken the matter up with the United States Department of Agriculture, through Senators Newlands and Nixon of the State of Nevada.

He wishes to secure from your department advices as to the conditions that exist in California in relation to diseases in animals and the advisability of continuing or revoking the quarantine which he has issued against California cattle.

While I do not wish to be considered as trying to influence your reply to Governor Oddie, I, however, think that California should be heard in the premises, and it is for this reason I am now writing you.

It appears that the primary reason for this petition was on account of the fact that Miller & Lux of this State had rented feed in Mono County in California, and wished to make several shipments of cattle from Buttonwillow, in Kern County, California, into Mono County, California, and in order to do this the shipment would have to be routed to Wabuska, in the State of Nevada, and from there driven into Mono County, California.

Nevada cattlemen, in conversation with me, termed Kern County "Southern California" and stated that fever existed there among cattle and it was dangerous to Nevada cattle for such Kern County cattle to come into, or be driven through any part of the State of Nevada.

The fever they refer to is undoubtedly Texas fever. However, in Governor Oddie's proclamation he also mentioned the following diseases: anthrax, blackleg and tuberculosis. I attach herewith copy of this proclamation.

On investigation I find further that Nevada cattlemen have been in the habit of securing this feed for their cattle in Mono County, California, and it might appear that the reason for this quarantine is due to a fight for cattle feed.

Newspaper dispatches from the State of Nevada also stated that it was thought that this quarantine was placed on California cattle as a reprisal for quarantines California placed on Nevada potatoes several months ago, on account of disease in these potatoes.

Now, as far as cattle ticks are concerned, it is unnecessary for me to inform you as to the conditions regarding same in California, inasmuch as you know the

only places where they exist at the present time are San Diego County and the western part of San Luis Obispo County and even in these sections rigid quarantines are enforced by the State and the eradication work is well in hand.

Anthrax has existed from year to year on a number of ranches in the San Joaquin and Sacramento valleys in this State. However, the disease only occurs during the fall months of the year; and during the past few years, through the efforts of this department, large cattlemen have kept their cattle off these infected fields during these months, so that the losses from this disease have been comparatively small.

Tuberculosis is, of course, prevalent more or less generally among our dairy cattle, as it is among the dairy cattle in nearly all the states in this country. Among range cattle it is comparatively rare, with the exception of a few herds.

If your department would attempt to personally investigate the above conditions in California, it would require a long time for you to do so, and you would obtain no further information than I have given you in the foregoing part of this letter.

Now, for the purpose of showing you the injustice of the quarantine issued by Governor Oddie, I wish to inform you, or remind you, of the conditions that exist in the State of Nevada, as far as sanitary police control in animals is concerned. In the State of Nevada anthrax is prevalent among cattle located on the lands along the Truckee, Carson and Humboldt rivers. Blackleg is a constant visitor among calves on the ranges of that State. Tuberculosis exists among cattle in the State of Nevada.

These are the three diseases, in addition to Texas fever, mentioned in Governor Oddie's proclamation. These three diseases, as you know, exist among live stock in many states of the United States. However, Nevada is one of the few states where absolutely no attempt is made to control or eradicate such infections.

Any attempt on my part to comment upon the incongruity of Governor Oddie's quarantine would therefore be ridiculous.

The State of California is not seriously affected by such a quarantine, inasmuch as only a few shipments are being offered to go into the State of Nevada. However, many cattlemen of this State have become unduly excited and are calling for reprisals, in that they wish this State to quarantine against the State of Nevada. This we could do with better grace and with more reason than Nevada's quarantine against California. However, we do not wish to resort to such measures, as this department believes that in so doing trade relations with Nevada would be seriously affected, inasmuch as great numbers of Nevada cattle and sheep come into California for summer feed; and Nevada's markets for horses, as well as for fat cattle and sheep, are situated within the State of California.

I would therefore request that if this matter is offered to you for a solution that you will give every phase of it the attention it demands and that in making your decision you be guided only by the broad principles involved.

Very respectfully yours,

[Signed] CHARLES KEANE,
State Veterinarian.

[Enclosure]

Later, the following reply was received from the United States Bureau of Animal Industry, Washington, D. C.:

MAY 31, 1912.

Dr. Charles Keane, State Veterinarian, Sacramento, Cal.

DEAR DOCTOR: Referring to your letter of the 25th instant and previous correspondence respecting the proclamation by Governor Oddie forbidding the admission of California cattle, I take pleasure in informing you that on the 27th instant the Secretary of Agriculture addressed a communication to Hon. Geo. S. Nixon, United States Senator, in effect as follows:

Referring again to your letter of the 16th instant, with inclosure from Messrs. Humphrey & Moffat, Reno, Nevada, respecting the proclamation by Governor Oddie forbidding the admission of California cattle, I desire to inform you that the inspector of this department who was directed to investigate conditions in California has submitted his report.

From the inspector's report it appears that while the diseases mentioned in the Governor's proclamation do exist to some extent in southern California as elsewhere, yet there have been no recent outbreaks. Texas fever infection is con-

fined to San Diego County and the western portion of San Luis Obispo County, each infested herd being under surveillance and this disease in California is under absolute control.

Very respectfully,

A. M. FARRINGTON,
Acting Chief of Bureau.

That the Governor of Nevada had been improperly advised concerning the health of California live stock, is very clear. However, as soon as the Governor ascertained the true facts, he revoked this quarantine, which was done on June 29, 1912, just one and one half months after its issuance.

SOUTHERN CATTLE FEVER QUARANTINE REGULATIONS ISSUED DURING THE SIXTY-SECOND AND SIXTY-THIRD FISCAL YEARS.

STATE OF CALIFORNIA,
OFFICE OF STATE VETERINARIAN,
SACRAMENTO.

PROCLAMATION.

EXECUTIVE DEPARTMENT, STATE OF CALIFORNIA.
SACRAMENTO, CAL., October 16, 1911.

WHEREAS, The State Veterinarian of the State of California has ascertained that cattle located in certain counties and portions of counties in the State of California hereinafter named are liable to communicate an infectious disease known as Texas, splenetic or southern fever to cattle located in other counties and portions of counties in the State of California should said cattle located in the counties and portions of counties hereinafter named be shipped, moved, transported, driven or grazed over the lands situated in other counties and portions of counties in the State of California; and,

WHEREAS, Under and by virtue of an act of the legislature of the State of California, entitled, "An act to protect domestic live stock from contagious and infectious diseases, to provide for the appointment and duties of officials to carry into effect the provisions of this act, and to provide an appropriation therefor," which became a law March 18, 1899, and as amended March 20, 1905, and March 23, 1907, and as further amended and approved March 19, 1909, the State Veterinarian of the State of California, in order to prevent the spreading or communication of said disease of Texas, splenetic or southern fever in cattle has, on this sixteenth day of October, 1911, quarantined the following counties and portions of counties in the State of California:

The entire county of San Diego.

The entire county of Orange.

All that portion of San Luis Obispo County lying west and south of a line commencing at the point of the intersection of the township line between township 32 S., R. 17 E. and 32 S., R. 18 E., and the Santa Maria River on the southern boundary of the county of San Luis Obispo, and extending northerly on said township line between townships 32 S., R. 17 E., 32 S., R. 18 E. and 31 S., R. 17 E., and 31 S., R. 18 E., to its intersection with the summit of the Santa Lucia range of mountains; thence following the summit of the Santa Lucia range of mountains northerly and northwesterly to its intersection with the northerly boundary line of San Luis Obispo County.

WHEREAS, The State Veterinarian has, on this sixteenth day of October, 1911, made and established the following rules and regulations as to the movements of cattle from said quarantined counties and quarantined portions of counties into other counties or portions of counties in the State of California:

From and after this sixteenth day of October, 1911, every person, company, corporation, their agents and servants, are hereby prohibited from driving, leading, moving, carrying or transporting, or from causing or permitting to be driven, led, moved, transported, carried or drifted into any other county or portion of county in the State of California, any cattle originating or being in the aforesaid quarantined counties and portions of quarantined counties, unless such cattle have first been inspected by the State Veterinarian or his duly authorized deputy and are accompanied by a certificate issued by such officer stating that said cattle are free from contagious and infectious diseases and cattle ticks (*Margaropus annulatus*); provided, however, that when said cattle are shipped or transported on railroad trains for immediate slaughter, inspection and certification of such cattle will not be required, but that when so shipped or transported by railroad the following regulations must be observed:

On unloading said cattle at their destination, or for food and rest en route to same, separate pens must be set apart to receive them and no other cattle shall be admitted to said pens. The cars that have carried said cattle shall be cleaned

and disinfected before they are again used to transport, store or shelter animals or merchandise.

All cars carrying said cattle shall bear placards stating that said cars contain "SOUTHERN CATTLE," and each of the way bills of said shipments shall have a note on its face with a similar statement.

The cars used to transport such cattle shall be cleaned and disinfected in the following manner:

(a) Remove all litter and manure. This litter and manure must be stored where no cattle can come into contact with it for a period of at least eight months.

(b) Wash the cars with water until clean.

(c) Saturate the entire inner surface of the cars, including the inner surface of the car doors, with a mixture made of one and one half pounds of lime and one quarter pound of pure carbolic acid to each gallon of water; or with a solution made by dissolving four ounces of chloride of lime to each gallon of water.

It is further ordered that a violation of any or either of the foregoing rules and regulations shall be an offense, and punishable as provided by the laws of the State of California.

NOW, THEREFORE, I, Hiram W. Johnson, as Governor of the State of California, by virtue of the authority vested in me by law, do hereby proclaim the counties and portions of counties named in this proclamation quarantined within the State of California, and further hereby proclaim the foregoing rules and regulations prescribed by the State Veterinarian for the maintenance and enforcement of such quarantine to be legal and binding rules and regulations within the State of California, and I do further proclaim that said rules and regulations shall be maintained and enforced within the State of California, and that a violation thereof shall subject all persons so violating any of said rules or regulations to the penalties provided for in section eight of that said act of the legislature of the State of California entitled, "An act to protect domestic live stock from contagious and infectious diseases, to provide for the appointment and duties of officials to carry into effect the provisions of this act, and to provide an appropriation therefor,"

In witness whereof, I have hereunto set my hand and caused the Great Seal of this State to be hereunto affixed, this sixteenth day of October, A. D., 1911.

[Signed] HIRAM W. JOHNSON,
Governor of the State of California.

[SEAL]

Attest:

[Signed] FRANK C. JORDAN,
Secretary of State.

STATE OF CALIFORNIA,
OFFICE OF STATE VETERINARIAN,
SACRAMENTO.

PROCLAMATION.

EXECUTIVE DEPARTMENT, STATE OF CALIFORNIA.
SACRAMENTO, CAL., March 7, 1912.

WHEREAS, The State Veterinarian of the State of California has ascertained that cattle located in certain counties and portions of counties in the State of California hereinafter named are liable to communicate an infectious disease known as Texas, splenic or southern fever to cattle located in other counties and portions of counties in the State of California should said cattle located in the counties and portions of counties hereinafter named be shipped, moved, transported, driven or grazed over the lands situated in other counties and portions of counties in the State of California; and,

WHEREAS, Under and by virtue of an act of the legislature of the State of California, entitled, "An act to protect domestic live stock from contagious and infectious diseases, to provide for the appointment and duties of officials to carry into effect the provisions of this act, and to provide an appropriation therefor," which became a law March 18, 1899, and as amended March 20, 1905, and March 23, 1907, and as further amended and approved March 19, 1909, the State Veterinarian of the State of California, in order to prevent the spreading or communication of said disease of Texas, splenic or southern fever in cattle has, on this seventh day of

March, 1912, quarantined the following counties and portions of counties in the State of California:

The entire county of San Diego.

All that part of Orange County located south and west of a line beginning at the Pacific Ocean in the town of Newport Beach; thence northeasterly following the center of the Newport-Tustin road to its intersection with the Santa Ana-Trabuco road; thence southeasterly following the center of said Santa Ana-Trabuco road to its intersection with the Aliso Canyon road; thence northeasterly following the center of said Aliso Canyon road to the north line of township 6 south, range 6 west; thence east on said township line to its intersection with the eastern boundary line of Orange County.

All that part of San Luis Obispo County lying west and south of a line commencing at the point of the intersection of the township line between township 32 S., R. 17 E. and 32 S., R. 18 E., and the Santa Maria river on the southern boundary of the county of San Luis Obispo, and extending northerly on said township line between townships 32 S., R. 17 E., 32 S., R. 18 E. and 31 S., R. 17 E., and 31 S., R. 18 E., to its intersection with the summit of the Santa Lucia range of mountains; thence following the summit of the Santa Lucia range of mountains northerly and northwesterly to its intersection with the northerly boundary line of San Luis Obispo County.

WHEREAS, The State Veterinarian has, on this seventh day of March, 1912, made and established the following rules and regulations as to the movements of cattle from said quarantined counties and quarantined portions of counties into other counties or portions of counties in the State of California;

From and after this 7th day of March, 1912, every person, company, corporation, their agents and servants, are hereby prohibited from driving, leading, moving, carrying or transporting, or from causing or permitting to be driven, led, moved, transported, carried or drifted into any other county or portion of county in the State of California, any cattle originating or being in the aforesaid quarantined counties and portions of quarantined counties, unless such cattle have first been inspected by the State Veterinarian or his duly authorized deputy and are accompanied by a certificate issued by such officer stating that said cattle are free from contagious and infectious diseases and cattle ticks (*Margaropus annulatus*); provided, however, that when said cattle are shipped or transported on railroad trains for immediate slaughter, inspection and certification of such cattle will not be required, but that when so shipped or transported by railroad the following regulations must be observed:

On unloading said cattle at their destination, or for food and rest en route to same, separate pens must be set apart to receive them and no other cattle shall be admitted to said pens. The cars that have carried said cattle shall be cleaned and disinfected before they are again used to transport, store or shelter animals or merchandise.

All cars carrying said cattle shall bear placards stating that said cars contain "SOUTHERN CATTLE," and each of the way bills of said shipments shall have a note on its face with a similar statement.

The cars used to transport such cattle shall be cleaned and disinfected in the following manner:

(a) Remove all litter and manure. This litter and manure must be stored where no cattle can come into contact with it for a period of at least six months.

(b) Wash the cars with water until clean.

(c) Saturate the entire inner surface of the cars, including the inner surface of the car doors, with a mixture made of one and one half pounds of lime and one quarter pound of pure carbolic acid to each gallon of water; or with a solution made by dissolving four ounces of chloride of lime to each gallon of water.

It is further ordered that a violation of any or either of the foregoing rules and regulations shall be an offense, and punishable as provided by the laws of the State of California.

Now, therefore, I, Hiram W. Johnson, as Governor of the State of California, by virtue of the authority vested in me by law, do hereby proclaim the counties and portions of counties named in this proclamation quarantined within the State of California, and further hereby proclaim the foregoing rules and regulations prescribed by the State Veterinarian for the maintenance and enforcement of such quarantine to be legal and binding rules and regulations within the State of California, and I do further proclaim that said rules and regulations shall be maintained

and enforced within the State of California, and that a violation thereof shall subject all persons so violating any of said rules and regulations to the penalties provided for in section eight of that said act of the legislature of the State of California entitled, "An act to protect domestic live stock from contagious and infectious diseases, to provide for the appointment and duties of officials to carry into effect the provisions of this act, and to provide an appropriation therefor."

In witness whereof, I have hereunto set my hand and caused the Great Seal of this State to be hereunto affixed, this seventh day of March, A. D. 1912.

[Signed] HIRAM W. JOHNSON,
Governor of the State of California.

[SEAL]

Attest:

[Signed] FRANK C. JORDAN,
Secretary of State.

STATE OF CALIFORNIA,
OFFICE OF STATE VETERINARIAN,
SACRAMENTO.

PROCLAMATION.

EXECUTIVE DEPARTMENT, STATE OF CALIFORNIA,
SACRAMENTO, CAL., March 7, 1912.

WHEREAS, The fact has been determined by the State Veterinarian of the State of California that an infectious disease known as splenetic, Southern or Texas fever exists among cattle in the following named states, to wit: Oklahoma, Texas, Missouri, Arkansas, Louisiana, Mississippi, Tennessee, Alabama, Virginia, North Carolina, South Carolina, Georgia and Florida; and

WHEREAS, Under and by virtue of an act of the legislature of the State of California, entitled, "An act to protect domestic live stock from contagious and infectious diseases, to provide for the appointment and duties of officials to carry into effect the provisions of this act, and to provide an appropriation therefor," which became a law March 18, 1899, and as amended March 20, 1905, and March 23, 1907, and as further amended and approved March 19, 1909, the State Veterinarian of the State of California, in order to prevent the spreading or communication of said disease of splenetic, Southern or Texas fever to cattle within the State of California, should cattle from the aforesaid states be imported into the State of California, has, on this 7th day of March, 1912, made and established the following rules and regulations as to the importations of cattle from the aforesaid states:

From and after this 7th day of March, 1912, every person, company, corporation, their agents and servants, are hereby prohibited from bringing into the State of California any cattle originating or being in that area of territory in the United States south of the Federal quarantine line which has been or may be established by the United States Department of Agriculture, or any cattle, or other animals infested with the *Margaropus annulatus* tick, except in accordance with the regulations of the United States Department of Agriculture, or except when said cattle are shipped or transported on railroad trains for immediate slaughter, and when so shipped or transported by railroad, the following regulations must be observed:

On unloading said cattle at their destination, or for food and rest en route to same, separate pens must be set apart to receive them and no other cattle shall be admitted to said pens. The cars that have carried said cattle shall be cleaned and disinfected before they are again used to transport, store or shelter animals or merchandise.

All cars carrying said cattle shall bear placards stating that said cars contain "SOUTHERN CATTLE," and each of the way bills of said shipments shall have a note on its face with a similar statement.

The cars used to transport such cattle shall be cleaned and disinfected in the following manner:

(a) Remove all litter and manure. This litter and manure must be stored where no cattle can come into contact with it for a period of at least six months.

(b) Wash the cars with water until clean.

(c) Saturate the entire inner surface of the cars, including the inner surface of the car doors, with a mixture made of one and one half pounds of lime and one quarter pound of pure carbolic acid to each gallon of water; or with a solution made by dissolving four ounces of chloride of lime to each gallon of water.

It is further ordered that violation of any or either of the foregoing rules and regulations shall be an offense, and punishable as provided by the laws of the State of California.

Now, therefore, I, Hiram W. Johnson, as Governor of the State of California, by virtue of the authority vested in me by law, do hereby proclaim the foregoing rules and regulations prescribed by the State Veterinarian to be legal and binding rules and regulations within the State of California, and I do further proclaim that said rules and regulations shall be maintained and enforced within the State of California, and that a violation thereof shall subject all persons so violating any of said rules and regulations to the penalties provided for in section eight of that said act of the legislature of the State of California, entitled, "An act to protect domestic live stock from contagious and infectious diseases, to provide for the appointment and duties of officials to carry into effect the provisions of this act, and to provide an appropriation therefor."

In witness whereof, I have hereunto set my hand and caused the Great Seal of this State to be hereunto affixed, this seventh day of March, A. D. 1912.

[Signed] HIRAM W. JOHNSON,
Governor of the State of California.

[SEAL]

Attest:

[Signed] FRANK C. JORDAN,
Secretary of State.

Respectfully submitted.

CHARLES KEANE,
State Veterinarian.

November 25, 1912.

MAP OF CALIFORNIA.



